

Air Travel Connectivity: A Global and Regional Network Analysis

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1 Introduction

In our analysis, we aim to tackle the field of air traffic, which concerns crucial aspects of our society such as international relations, military, tourism and, most importantly, trade. These are all immensely important, and rely on the assumption that the air traffic network is suitable, reliable and resilient. By deploying the techniques studied in the Social Network Analysis course, we aim to identify the key elements of such an important network, as well as underlining its limits.

2 Problem and Motivation

Air transportation has a crucial role in global connectivity, not only in the context of tourism, but also in facilitating trade and international relations. Therefore, we believe that it is extremely important to understand the efficiency of the network, in order to optimize operations, identifying strategic hubs, improve passenger's satisfaction and ensure resilience against disruptions (i.e. Natural disasters). Social Network Analysis can provide a powerful framework for examining the connectivity of airports and flights, as well as evaluating the efficiency of the network.

Another goal of our work (inspired by "*Aviation diplomacy: relationship between aviation and international relations*" Michał Marcin Kobiński, 2020 [1]) is analyzing the possibility of inferring political relations and social structures through the observation of flights within and between countries.

The main contributions that our research can provide, therefore, are:

1. identification of key hubs, so those airports which have a very high connectivity
2. detection of clusters of highly connective airports
3. relation between air traffic and political structures

3 Datasets

For this project we combine the informations available on two different datasets that we found on Kaggle [2] to access knowledge about airports [3] and airline routes [4] worldwide.

Besides the two databases, we relied on external data in order to (i) link each country with the respective continent [6], (ii) provide the economic data [8] necessary for the analysis that will be explored in the dedicated chapter of this report.

There was no need for digitization, as the data is already structured in a digital format of cvs files. The data is also publicly accessible under Open Database license [5], which makes it a reliable source, free of licensing complications.

The obtained network is directed, with weighted edges (representing flight frequency between airports) and monomodal (nodes = airports) but incorporates heterogeneous node attributes (continent and country), enabling region-specific observations.

4 Validity and Reliability

The Routes Database [4], comprising 59,036 routes across 3,209 airports and 531 airlines globally, provides a substantial foundation for modeling flight networks, ensuring a proportional representation of flight distributions across countries, airports, and geographic regions. While real-world air traffic encompasses a significantly larger volume of flights, computational constraints and access to data necessitated the use of a smaller dataset. Notably, the dataset's global coverage ensures that all major geographic areas are included, enhancing its validity as a representative model of real-world aviation networks.

Regarding reliability, the absence of data preprocessing and the use of linear, transparent methodologies to construct the network graph ensure reproducibility. The straightforward transformation of raw route data into nodes (airports) and edges (routes) allow other researchers to replicate the analysis with minimal ambiguity.

5 Measures and Results

In this section, we outline the measures applied to the network and their utility in this specific topic. To delimit the scope of potential observations, we focused on Africa, the USA, Europe, and the World to consider diverse political and geographic contexts. National and continental subnetworks were constructed by considering only nodes and edges internal to each geographic subnetwork's area.

5.1 Centrality Measures

5.1.1 Degree Centrality

Weighted degree centrality reflects each airport's total traffic volume. Globally, Europe, Africa, and the United States are dominated by a few major hubs with extensive, high-volume connections, leaving most airports with significantly lower values - such as BCN (Barcelona). This results in a heavily skewed distribution, underscoring the network's reliance on a small subset of traffic-intensive nodes.

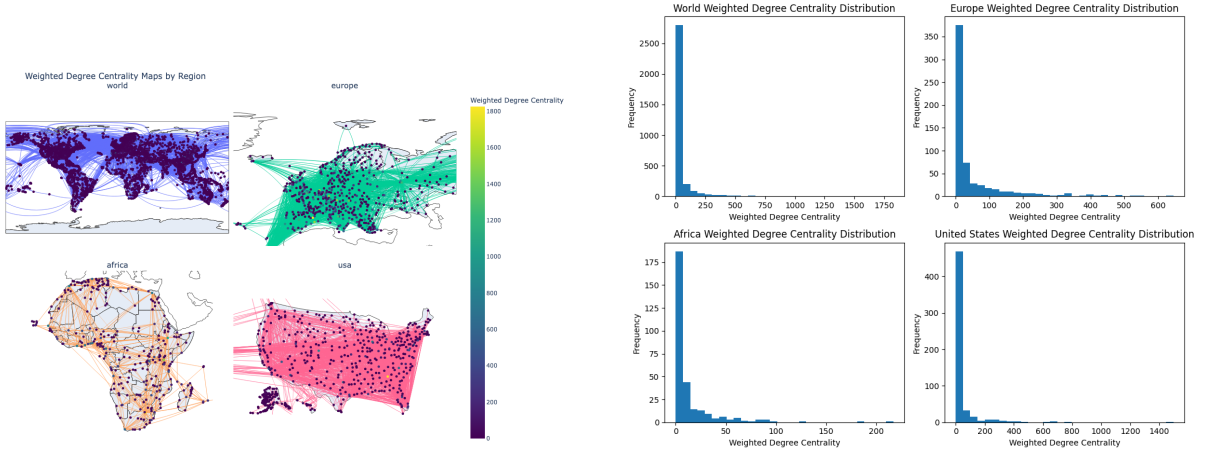


Figure 1: Degree centrality (weighted)

5.1.2 Eigenvector Centrality

Weighted eigenvector centrality measures not only an airport's own traffic but also that of its influential neighbors. Again, each region's busiest airports consistently rank highest, with some U.S. hubs standing out among the global top. Africa tends to show more uneven patterns, where high-volume nodes may still be relatively isolated from other key airports. Overall, these findings highlight the role of a handful of major hubs that shape interconnectivity across the network.

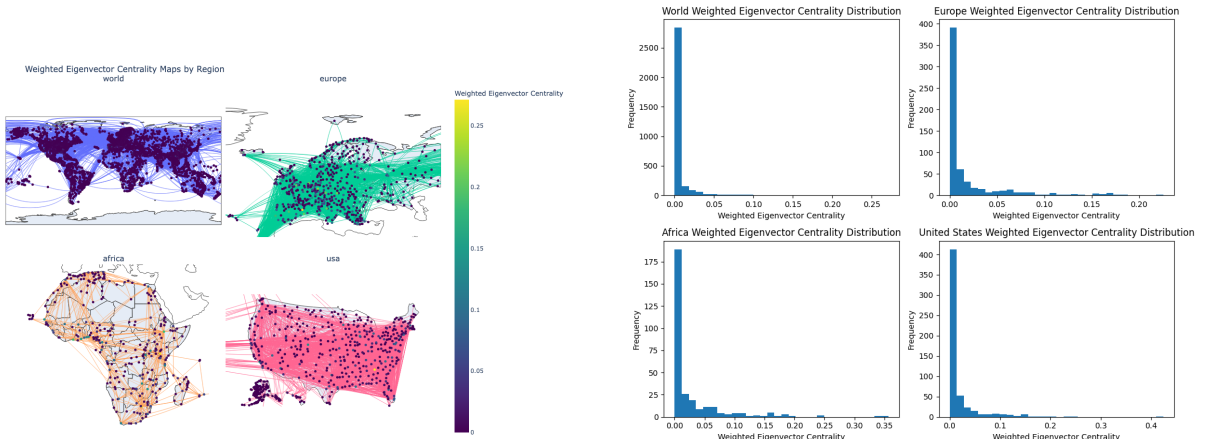


Figure 2: Eigenvector centrality

5.1.3 Closeness Centrality

Weighted closeness centrality measures how quickly an airport can reach others when accounting for flight volume or distance. Top-ranking airports in the United States and Europe benefit from well-developed, high-frequency routes, yielding higher accessibility. In contrast, African airports generally have lower closeness scores, suggesting fewer or more geographically dispersed connections. This uneven distribution underscores how localized infrastructure and travel demand shape regional accessibility.

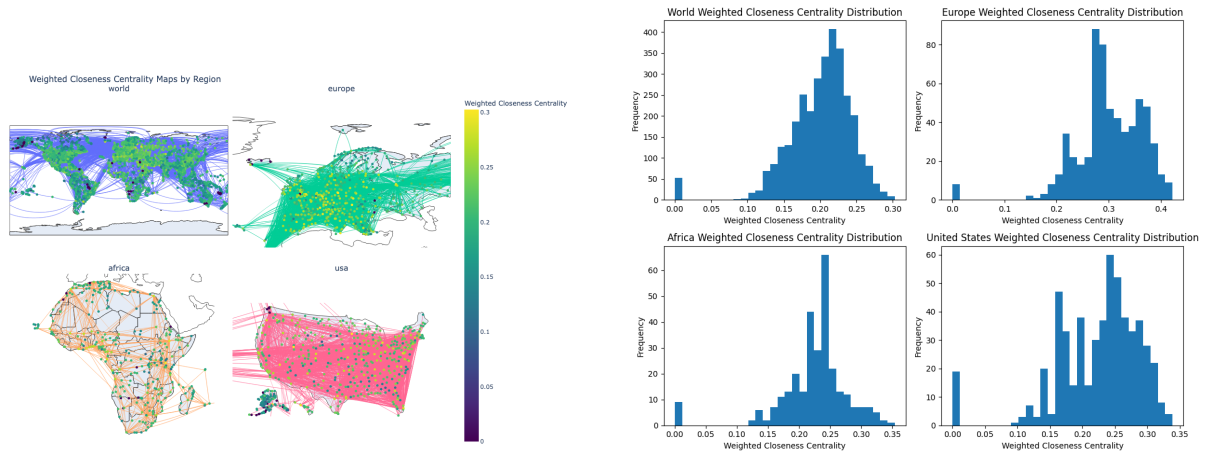


Figure 3: Closeness centrality

5.1.4 Betweenness Centrality

Weighted betweenness centrality highlights airports that frequently serve as high-traffic bridges for shortest paths. Across the global network, only a small fraction of airports attain substantial betweenness scores, while most remain negligible. In Africa, certain nodes stand out as key connectors for otherwise distant routes, whereas in the United States, a few critical airports dominate, reflecting dense traffic flows within and across the region.

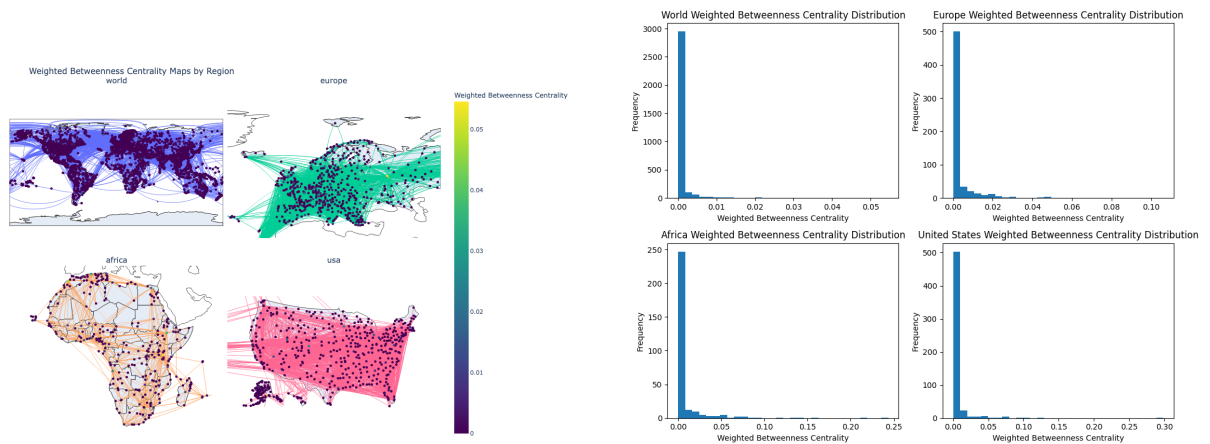


Figure 4: Betweenness centrality

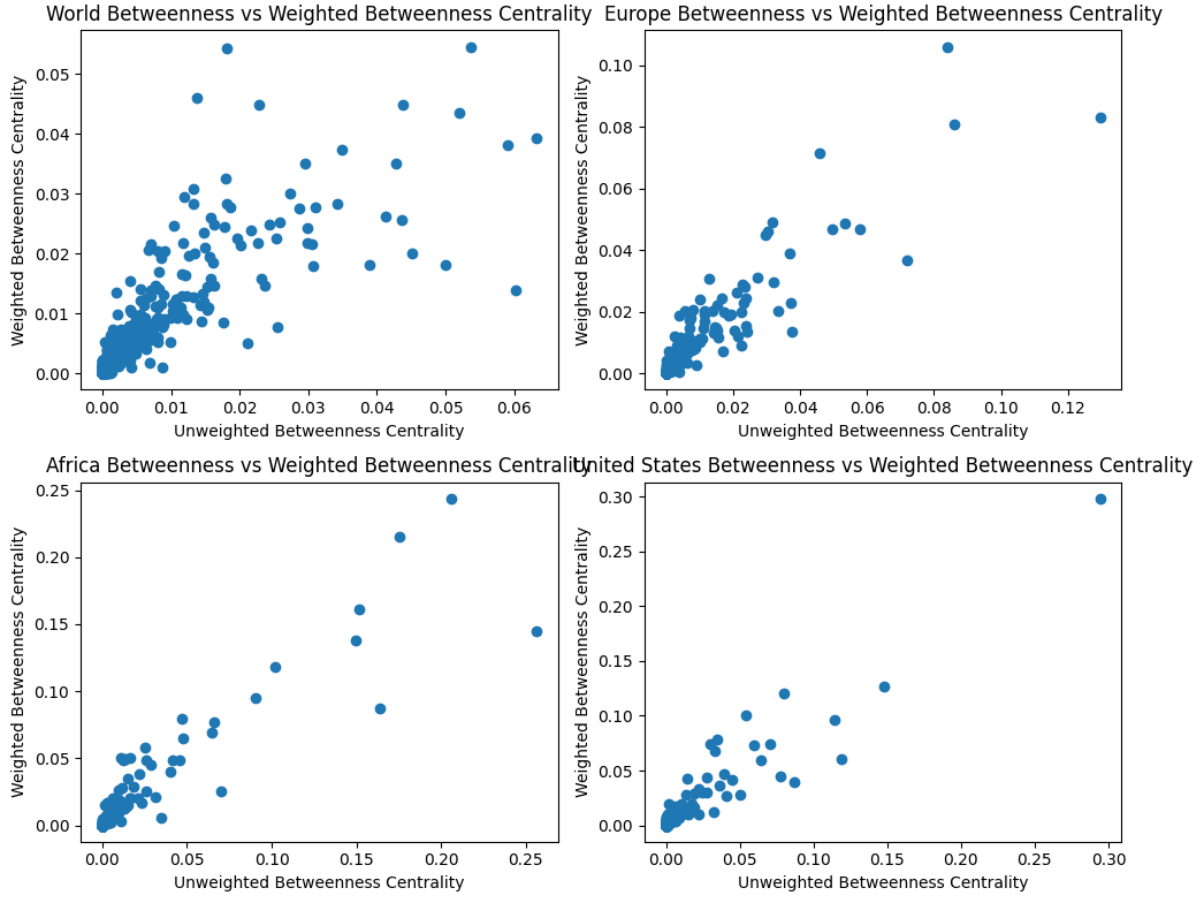


Figure 5: Betweenness vs Weighted Betweenness

5.1.5 Power Law and Log-Log Analysis of Centrality Distributions

Log-log plots and polynomial fittings suggest that weighted degree and weighted betweenness often come closest to following a power-law relationship, reflecting a scale-free structure where a small set of airports dominate. Weighted eigenvector distributions also show heavy-tailed tendencies but can deviate more, particularly in Africa. Weighted closeness tends to exhibit additional curvature or multi-peaked behavior, especially in Africa and, to a lesser extent, the United States, indicating geographic and infrastructural nuances. Overall, these heavy-tailed distributions reinforce the idea that global and regional air networks rely on a limited number of extremely influential hubs.

Centrality Distributions with Polynomial Fit (Log-Log)

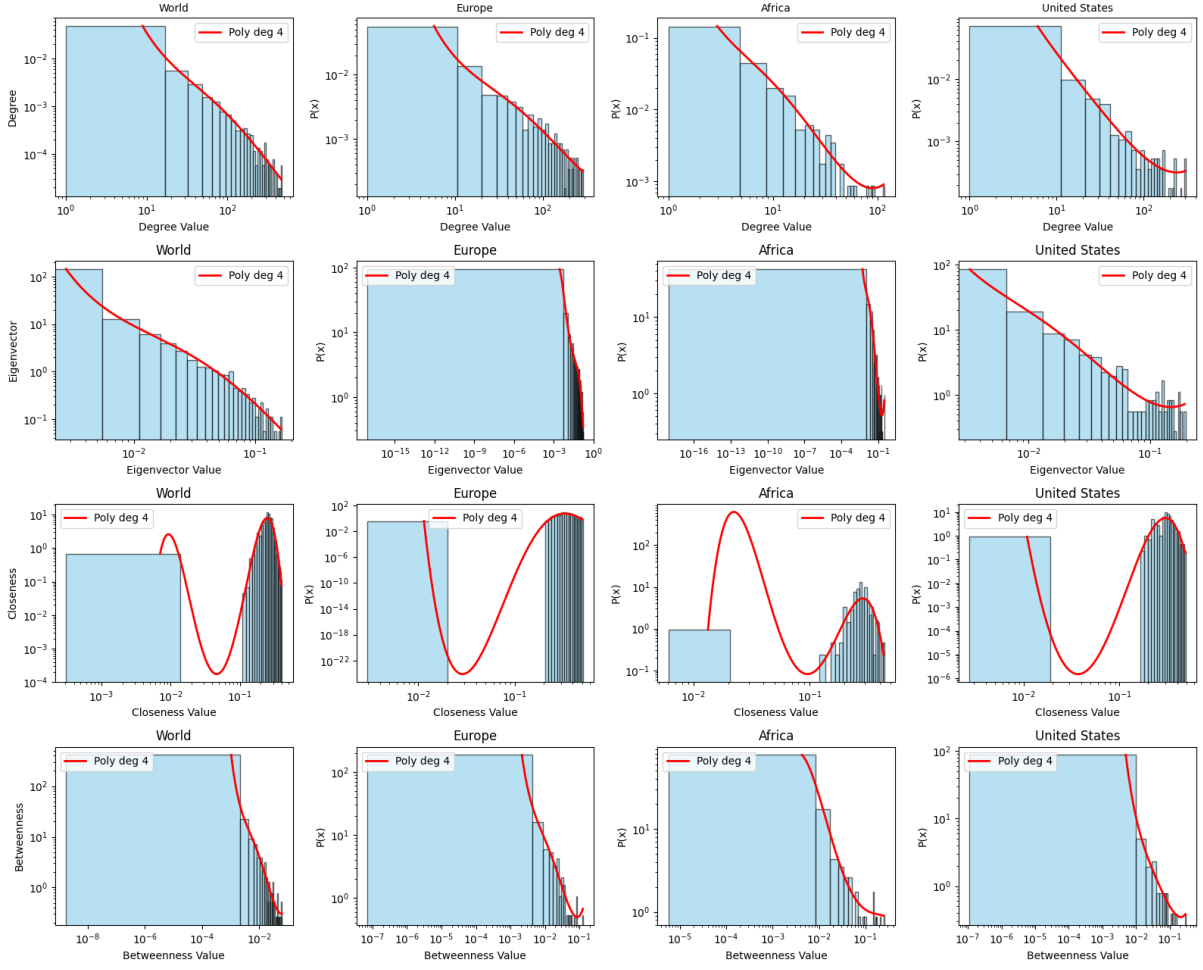


Figure 6: Centrality distribution logarithmic visualization

5.1.6 K-means Clustering

Applying K-means clustering in air traffic networks to the centrality metrics observed in the previous paragraphs can reveal critical structural insights: by classifying nodes based on their role in the network is possible to observe differences in the aviatory system of the observed regions.

- **Europe:** The European network exhibits numerous medium-centrality nodes (Cluster 1-2), reflecting its decentralized structure. However, these nodes gain *global* centrality when analyzed within the World network. This duality arises from Europe's political fragmentation, which creates multi-layered connectivity structure.
- **USA & Africa:** Both networks show polarization: few medium-centrality nodes, with traffic concentrated on high-centrality hubs (Cluster 3). In the USA, these hubs are coastal, aligning with economic density and frequency of transoceanic routes.

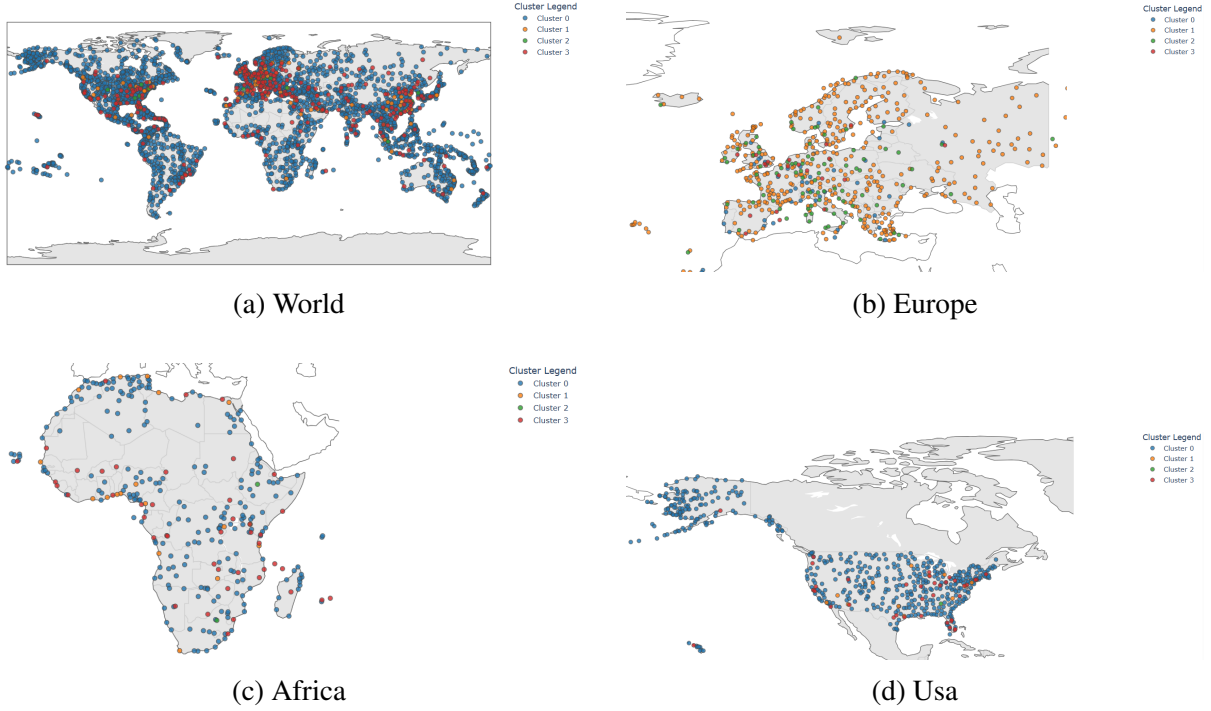


Figure 7: K-means clustering on centrality measures: higher cluster indices denote greater centrality (e.g., Cluster 3 = high centrality).

5.2 Homophily and Assortative Mixing

In the context of this study, assortativity was used to evaluate the propensity of airports (nodes) of a specific political entity to maintain flight connections with other airports belonging to the same political entity.

The assortativity between two regions (countries or continents) r_1 and r_2 is defined as:

$$\text{Assortivity}(r_1, r_2) = \underbrace{\frac{\#edges(r_1 \rightarrow r_2)}{m}}_{\text{Actual Fraction}} - \underbrace{\frac{\sum_{u \in r_1} \sum_{v \in r_2} d_{out}^u \cdot d_{in}^v}{m^2}}_{\text{Expected Fraction}},$$

where:

- $\#edges(r_1 \rightarrow r_2)$: Number of directed flights from region r_1 to r_2
- m : Total number of edges (flights) in the network
- d_{out}^u : Out-degree of node u (region r_1)
- d_{in}^v : In-degree of node v (region r_2)

Positive values of this difference indicate a higher number of flights between the regions than expected under the random model, whereas negative values suggest lower connectivity than expected.

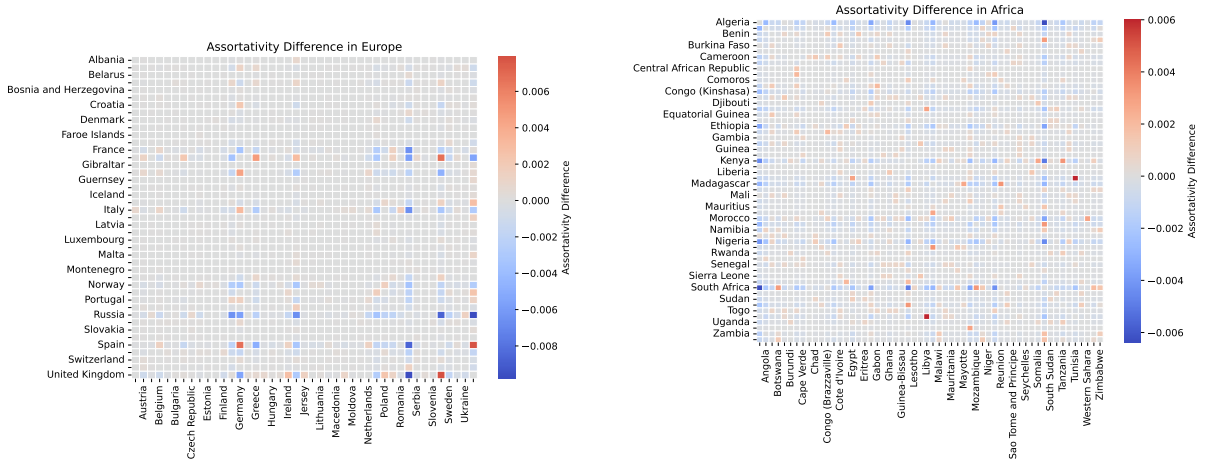


Figure 8: Assortativity patterns in European and African airports networks.

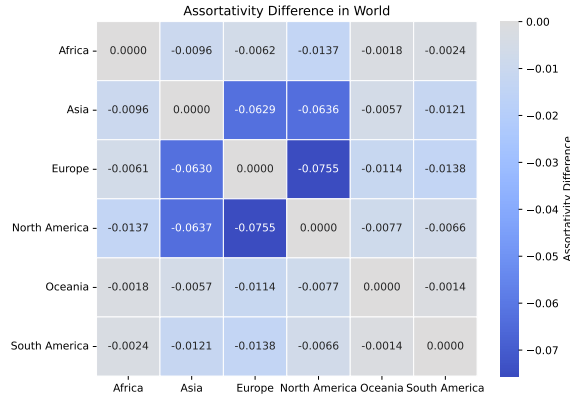


Figure 9: Assortativity patterns in intercontinental routes.

Results:

- The assortativity matrix, though not necessarily symmetric by definition, was empirically symmetric in our data. This implies that **asymmetry between two political regions** in terms of inbound and outbound flights is very **rare**.
- **Country-Level analysis in Europe and Africa:** The data show that high homophily between countries reflects both geographical proximity and political-historical ties. In Africa, for instance, the strong connectivity between Libya, Tunisia, and Egypt is an indicator of the deep historical and political links between north african nations. Similarly in Europe, high assortativity between Germany–Italy and UK–Spain likely mirrors significant tourism flows [10]. In contrast, Russia’s noticeable negative assortativity with Western European nations highlights the impact of the political isolation of this country in European politics.
- **Intercontinental analysis:** All continental pairs show negative assortativity, reflecting air traffic’s intra-continental dominance. Inter-continental flights are sparser than expected under random connectivity models, with far stronger negative assortativity (minimum 0.07 vs. 0.008 for intra-European/African country pairs). This highlights geography’s outsized role over politics: Europe–North America—historically politically

aligned—has the weakest connectivity, emphasizing how physical proximity drives flight networks more than diplomatic ties.

5.3 Average Local Clustering Coefficient by Node Degree

This analysis studies node degree vs. local clustering coefficient (LCC) for nodes with more than 2 neighbors. The directed flight graph was converted to undirected to clarify degree-LCC relationships and independent from flight direction.

The local clustering coefficient (lcc) for node i with degree d_i has been calculated with `networkX.clustering` [9] as:

$$lcc_i = \frac{2T_i}{d_i(d_i - 1)}$$

Where:

- T_i = Number of triangles through node i
- $\binom{d_i}{2} = \frac{d_i(d_i - 1)}{2}$ (maximum possible triangles)

The analysis was performed over the World graph, and over the 3 subgraphs representing the internal flights inside Europe, Africa and Usa.

Results:

- **The general results** confirm the hypothesis of an inverse relationship between node degree and average local clustering coefficient. This pattern aligns with common structural properties of transportation networks.
- **Europe vs. World:** Europe's degree-LCC trend is the most correlated with the global graph due to Europe's politically fragmented structure. Unlike the centralized USA, Europe's aviation network resembles a microcosm of the world:
 - Its patchwork of sovereign states (e.g., France, Germany) creates semi-autonomous regional clusters, each with hubs (Paris, Amsterdam) bridging local and international routes.
 - Airports serve multi-layered roles: connecting local regions (Milan to Rome), neighboring countries (Berlin to Warsaw), and global hubs (London to Moscow).

This nested hierarchy mimics the global network, where continents act as "countries" and hubs bridge them. The USA's unified aviation policy lacks this fractal-like structure.

- **USA:** Low-degree US airports often show low LCC (unlike Europe), reflecting: (i) a hub-and-spoke system where small airports act as spokes, connecting only to hubs—not each other—creating star-shaped networks with few triangles, (ii) interstate highway investments that further reduced regional flight demand, prioritizing long-haul efficiency over local clustering.

The hub-and-spoke system in the US creates structural holes, as the lack of connections between regional spokes positions hubs as critical gatekeepers controlling flows between otherwise disconnected nodes.

- **Africa:** Smaller node degrees reflect hubs prioritizing intercontinental routes and economic barriers (poverty, visas) limiting regional demand. Despite lower maximum degrees, Africa maintains the inverse degree-LCC trend: intra-African hubs act as “miniature” global bridges, with consistently low clustering.

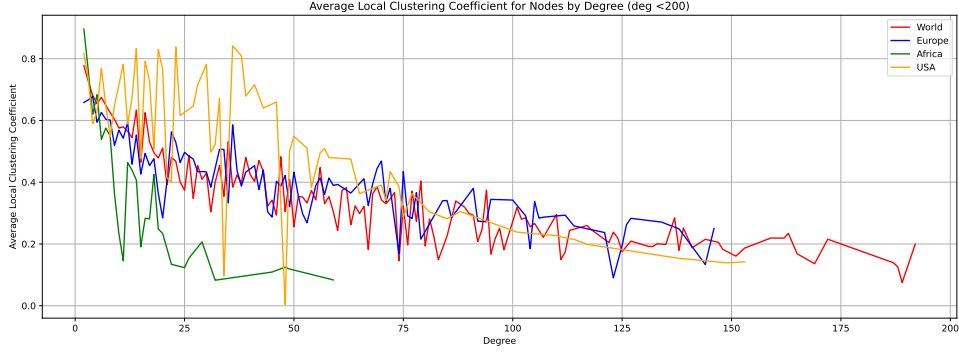


Figure 10: Local Clustering Coefficient for Usa, Africa, Europe, World

5.4 Analysis of Internal Flights and Network Density vs GDP

We aimed to make some observations regarding the relationship between the economy and the configuration of a country’s domestic air routes. The hypothesis is that it is possible to identify a correlation between these two aspects.

As a measure of economic performance, we considered the Nominal Gross Domestic Products (2024) [8].

The measures adopted were the number of internal flights and the density of the subgraph corresponding to a given country, defined as:

$$\text{density} = \frac{\frac{m}{\text{edges}}}{\frac{n}{\text{nodes}} \left(\frac{n}{\text{nodes}} - 1 \right)}$$

Results:

- **Strong correlation between internal flights and GDP:** Larger economies naturally require more extensive domestic air transport infrastructure to support economic activity.
- **Weak correlation between density and GDP:** Smaller nations may exhibit high density due to having fewer airports, whereas larger, highly urbanized nations might rely on alternative modes of transport such as railways (e.g., Japan). In addition to that, the internal structures of aviatory systems may express a low density for highly hub-reliant networks (e.g. Usa).

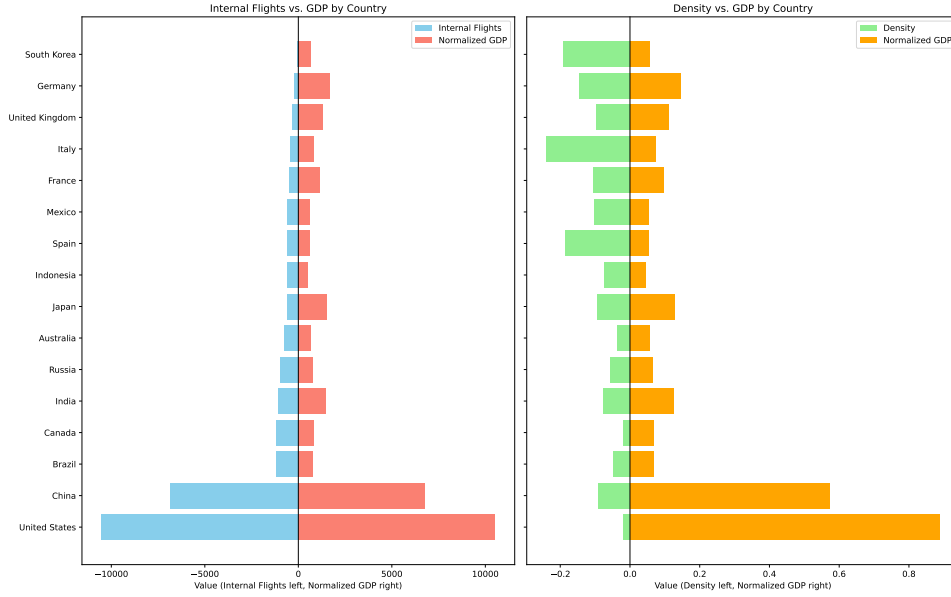


Figure 11: Flights Network properties vs GDP

5.5 Network Robustness Analysis in Iterative Hub Removal

Air routes provide an important indication of the crucial points in global logistics and transportation. It is therefore useful to observe the impact of large hubs on global connectivity. To do so we examined how susceptible connectivity would be in our observed regions of the world to the removal of main hubs.

In general, the analysis focuses on three key topological metrics:

- Density (defined in previous paragraph)
- Number of nodes in the largest strongly connected component
- Total number of strongly connected components

These measures are used to evaluate graph robustness when the most important hubs (detected with Hyperlink-Induced Topic Search [7]) are removed.

Results:

The global network, characterized by an initially very low density, exhibited a proportional reduction in density after hub removal, a smaller reduction compared to other networks. In contrast, the Usa network, which relies heavily on a centralized hub-and-spoke architecture, suffered the most pronounced disruption across all three metrics. This underscores its dependence on critical hubs for connectivity. Conversely, the European network, with the highest density, demonstrated remarkable resilience in component-related metrics that remained relatively stable, reflecting its dense, decentralized structure. These findings highlight how network density and centralization shape resilience, with centralized systems (USA) being more vulnerable to hub loss than dense, distributed systems (Europe).

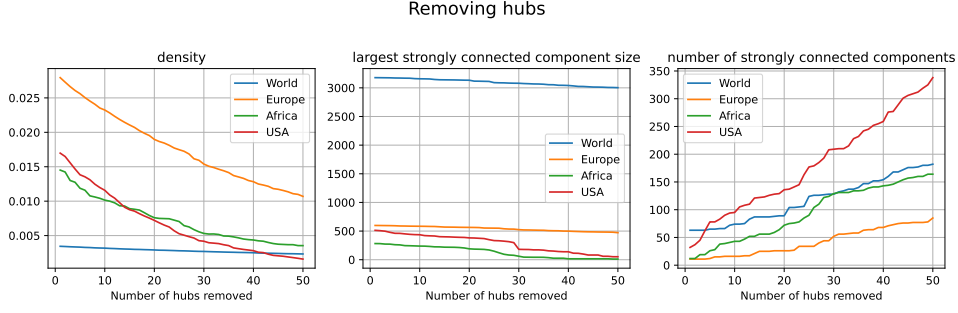


Figure 12: Network Robustness vs Hubs removal

5.6 Analysis of Reliance on Foreign Hubs

This analysis aims to identify countries that exhibit disproportionate reliance on foreign hubs for their network connectivity. To quantify this dependency, we compute for each country c a reliance ratio R_c defined as:

$$R_c = \frac{F_c}{T_c} \quad (1)$$

Where:

- F_c : Number of flights from airport in country c to top-ranked hubs located in foreign territories.
- T_c : Total number of flights to a foreign airport for all airports in country c .

Results:

The results confirm two primary categories of countries with excessive foreign hub reliance:

- Very small and/or island nations (e.g., Gibraltar, Belize, Iceland, Cayman Islands).
- Countries experiencing historical or political isolation (e.g., Iraq, North Korea).

This pattern demonstrates how limited domestic network infrastructure or geopolitical constraints manifest in measurable connectivity dependencies through the R_c metric.

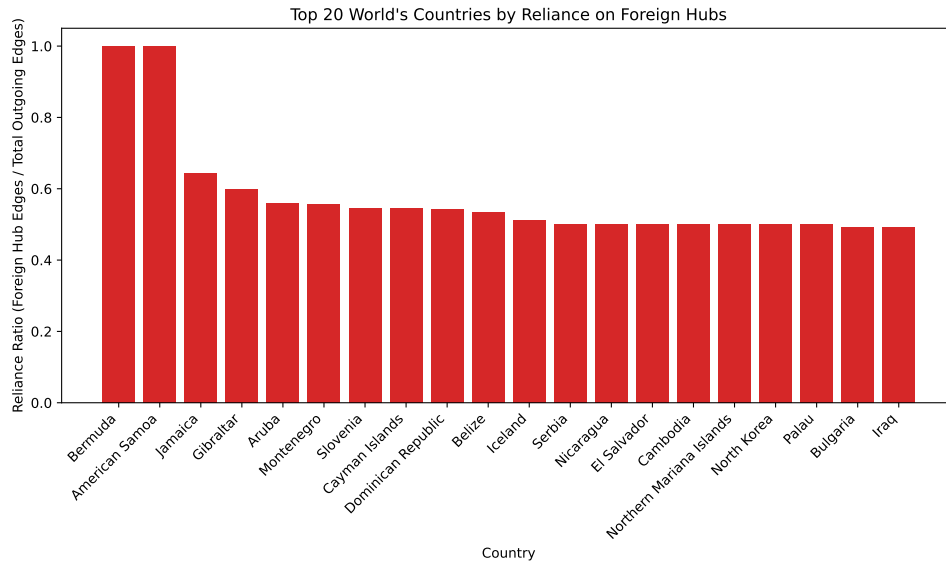


Figure 13: Countries with highest reliance ratio (R_c) worldwide

5.7 Cliques

Cliques are groups of nodes where every element is directly connected to every other in the set. This measure is particularly useful in our analysis, as identifying cliques reveals clusters of highly interconnected airports. These clusters often represent regional airline networks serving specific markets, helping us understand the structure of air traffic connectivity.

Results:

An interesting insight from this analysis is that the largest clique in the global airport network (consisting of 24 elements) is nearly identical to the one found when restricting the graph to only U.S. airports. The U.S.-only clique has 23 members, including: ['ATL', 'ORD', 'DTW', 'CLT', 'PHL', 'DCA', 'EWR', 'DFW', 'IAH', 'BOS', 'DEN', 'MSP', 'LAS', 'LAX', 'PHX', 'MCO', 'SFO', 'STL', 'FLL', 'MKE', 'MCI', 'MSY', 'CLE']. The only difference in the global version is the addition of Toronto Pearson Airport (YYZ) in Canada.

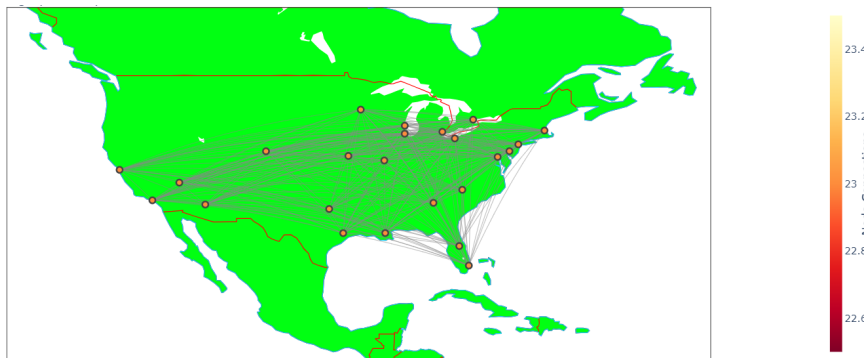


Figure 14: largest clique in the world context

The European continent behaves much like the United States, with a largest clique that involves 22 nodes, therefore there is not significant difference between these analyzed territories.

Africa's clique is surprisingly small for such a large continent, consisting of just seven airports: ['LFW', 'COO', 'LOS', 'ABJ', 'CMN', 'LBV', 'DLA'], all on the western coast. These airports are key hubs for trade and commerce, facilitating regional and international connections

that support economic activity in West and Central Africa. Still, the presence of a clique this small suggests low connectivity within continent.



Figure 15: largest clique in the African continent

5.8 K-Cores

This measure can be interpreted as a less strict version of a clique, as in this context a node needs to have K neighbors to be considered part of a core. The way that this is useful to our research is because it aids in identifying a densely connected nucleus of a network. This offers a more global view of air traffic, and which airports are crucial in ensuring connectivity.

Results:

In the world context, the largest K -core has $k = 162$ and contains 82 nodes. It involves mainly airports in Europe, but also includes airports like JFK (New York) and DXB (Dubai).

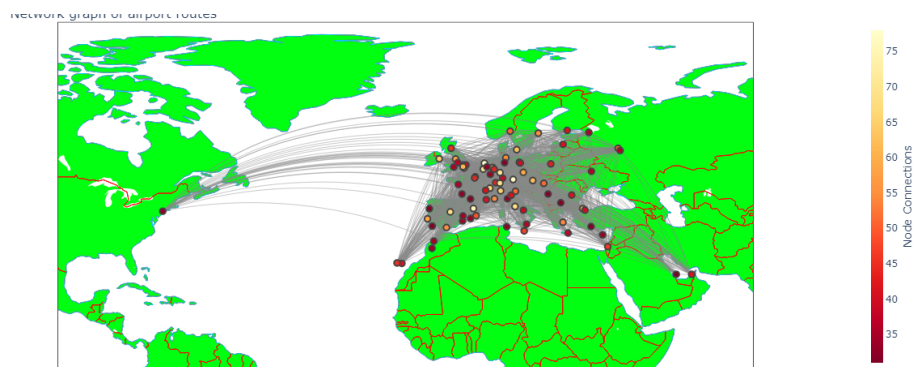


Figure 16: K-cores distribution in world territory

Furthermore, in the following plot we show the distribution of the cores across the k -cores in the network. This shows that most nodes belong to low- k cores (less than 8), meaning they have relatively low connectivity. The highest k -core (162) contains very few but extremely well-connected nodes, likely representing major network hubs.

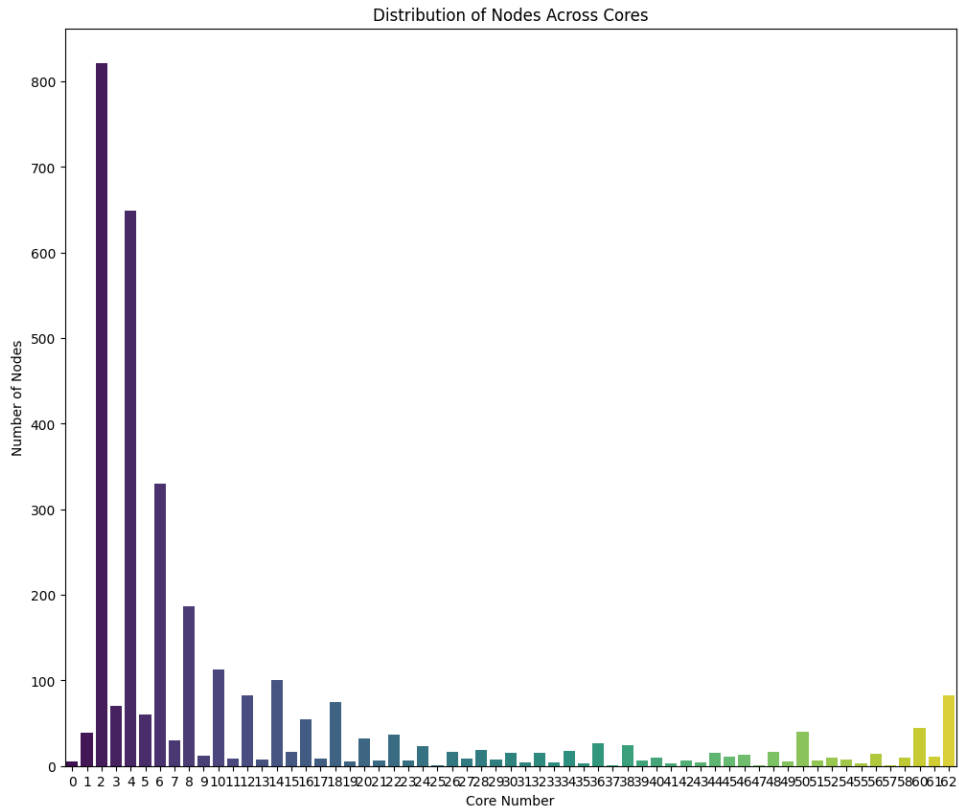


Figure 17: Distribution of nodes across K-cores

Europe shows extremely strong connectivity in this sense, as its k-core size is 66. In this case, the core density appears to be much stronger in central and western Europe, and we can observe a decline in connectivity towards the periphery of the core.

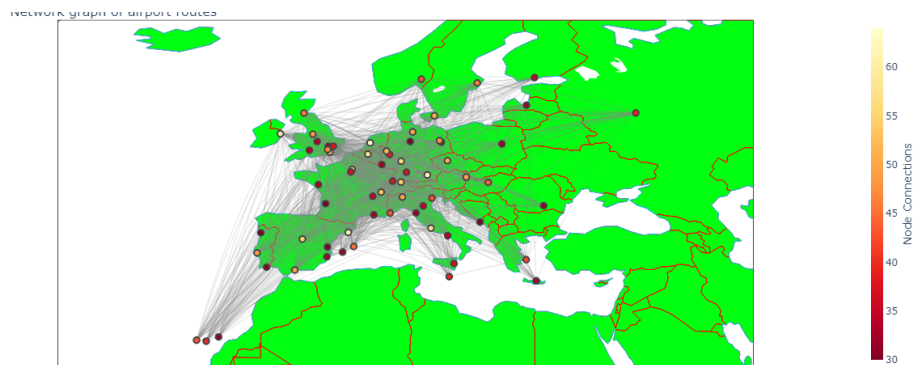


Figure 18: K-cores distribution in European territory

The African context remains an outlier, as observed in the clique analysis. In this case, the core size is 22, further reinforcing the claim of low connectivity across the region. The geographic distribution of core nodes is largely similar, with the highest concentration still found along the western coast. However, this analysis also reveals the presence of core nodes in the southern and eastern parts of the continent, indicating a slightly broader but still limited connectivity.

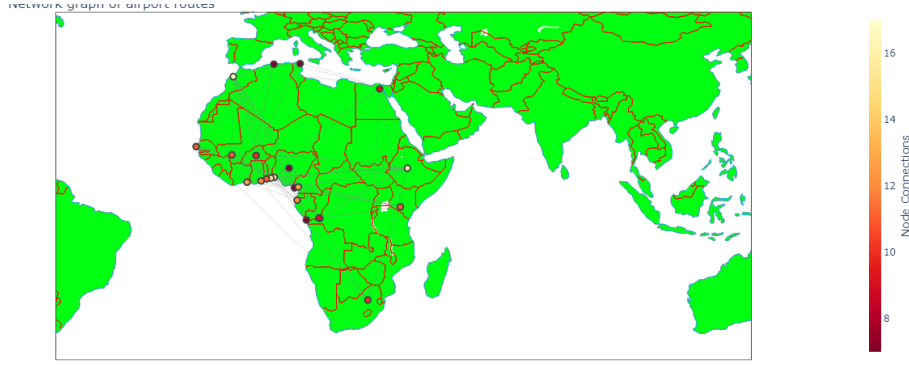


Figure 19: K-cores distribution in African territory

5.9 Component Analysis

A component is a group of nodes where each node is reachable from every other node within the set. This concept is crucial for analyzing network connectivity, as it helps determine whether a network is fully connected or fragmented into isolated subnetworks. For instance, these components can reveal remote regions, isolated islands, or distinct regional air networks, some of which may represent military or private connections.

In our analysis, we focused on Strongly Connected Components (SCCs) and Weakly Connected Components (WCCs). An SCC is a subgraph where every node can reach any other node within the component via a directed path. A WCC is a subgraph where, if we ignore edge directions, every node can still reach every other node within the component. This helps identify subnetworks that remain connected if flight directions are disregarded.

Results:

In our case, we have 52 SCCs, which means that the network in itself is not strongly connected. Furthermore, we have 12 WCCs. The largest WCC has 3,231 nodes, highlighting the extent of connectivity within this subnetwork.

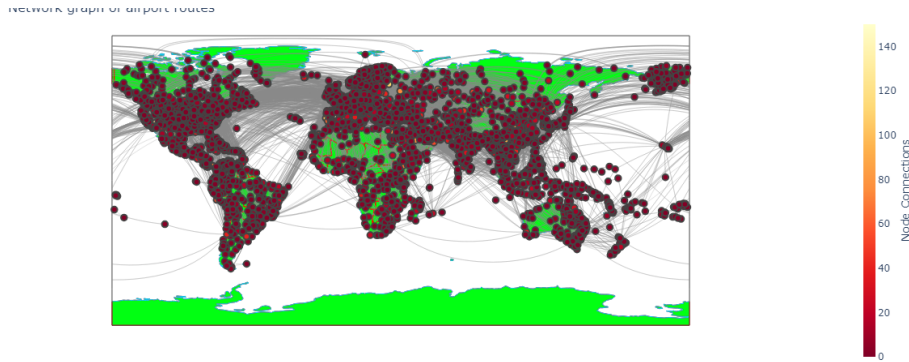


Figure 20: Largest Weakly Connected Component in the world

To break this down into the regions that we are analyzing:

- *Europe*: exhibits 7 SCCs and 6 WCCs, reflecting its dense and interconnected air transport network. The largest WCC connects 604 of the 612 nodes present in the European subgraph, which solidifies the claim that the European airport network is highly integrated.
- *USA*: In this territory we find 22 SCCs and 6 WCCs. This means that even if in a less

cohesive way than Europe, the network is still well connected, and out of the 552 total nodes, 541 belong to the same WCC.

- *Africa*: has 10 SCCs and 4 WCCs, which shows some level of fragmentation but still decent connectivity.

Overall, we can infer that in all the analyzed networks we can see some level of fragmentation, in some areas more than others, nonetheless the connectivity is always acceptable on a large scale.

5.10 Periphery Structure

The periphery consists of nodes that have the largest shortest path distances to others. This identifies those nodes which are located at the margins of the network but still contribute to connectivity. In our case, this underlines which are the most isolated airports, usually located in remote locations. This is of course useful for economic purposes, as it can show those locations where connectivity needs to be improved in order to enhance trade and tourism.

Results:

This measure was analyzed in the largest interconnected subset of the network, and from here we were able to find out that:

- **World**: the periphery nodes are ['BVI', 'CMP', 'XTG', 'FMI', 'THU', 'PTJ', 'STZ', 'YPO', 'YZG'], and they are almost all located in remote areas of the world like the British Virgin Islands (BVI) or Greenland (THU).

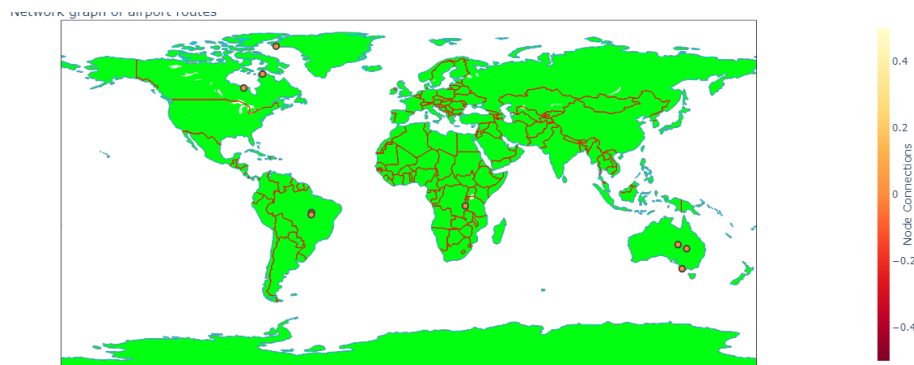


Figure 21: Periphery nodes in world context

- **Africa**: In this context, the peripheral nodes are situated in regions that were previously observed to be absent from the largest clique or k-core. This indicates that these areas of the continent have limited connectivity to the rest of the network.
- **Europe**: In the European continent, exactly as expected, the periphery nodes are located in all the parts which are harder to reach. We can find these nodes in the most northern part of the continent, close to the Arctic, as well as the islands and the central part of Russia.
- **USA**: The USA's periphery nodes are in remote areas like Alaska and the central deserts, while Hawaii, despite its remoteness, lacks them—likely due to its high tourism.

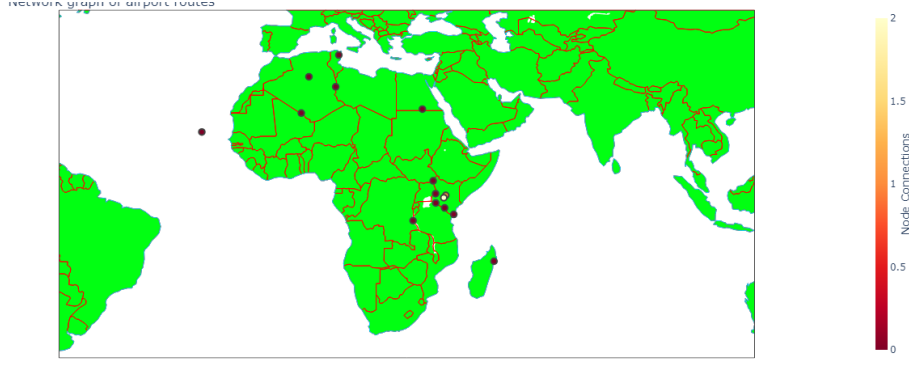


Figure 22: Periphery nodes in African context

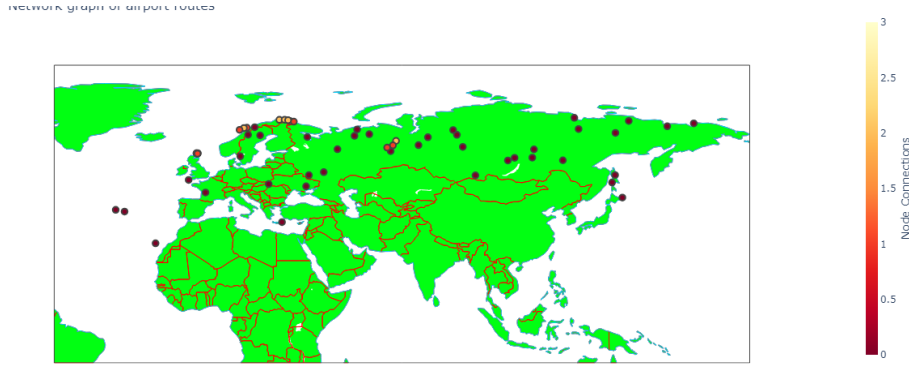


Figure 23: Periphery nodes in European context

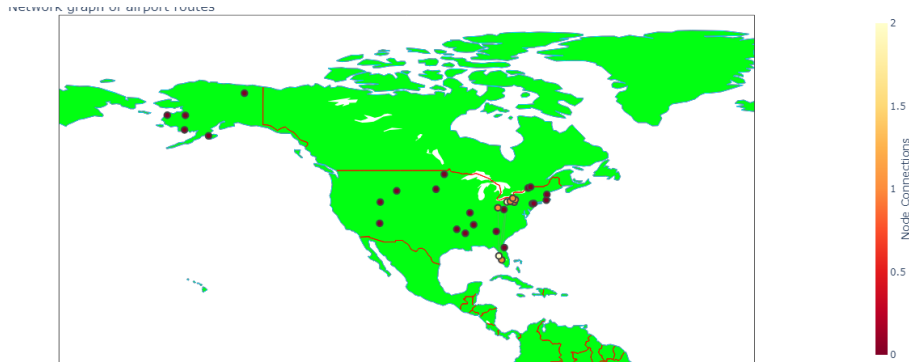


Figure 24: Periphery nodes in USA context

5.11 Small World Effect - Hubs and Authorities

This is one of the most interesting and important measures to analyze in this type of problem. Most real-world networks exhibit small-world properties, meaning that any node can be reached in a very limited number of steps from any other location in the network.

Mathematically, a network exhibits small-world properties if the mean shortest path length ℓ of the network is proportional to the logarithm of the number of nodes n :

$$\ell \propto \log n$$

where ℓ is defined as:

$$\ell = \frac{\sum_{ij} d_{ij}}{n^2}$$

with d_{ij} representing the shortest path length between nodes i and j . Referencing this concept, we have identified *hubs* and *authorities* within our network:

- *Hubs* are airports with the highest number of direct connections.
- *Authorities* are critical stopover points that facilitate connections between key locations.

Results:

Our analysis shows that hubs and authorities largely coincide. The main findings are:

- **World:** The network exhibits small-world properties as the average shortest path length is proportional to $\log(n)$. The top three airports serving as both hubs and authorities are ATL (Atlanta), LHR (London Heathrow), and ORD (Chicago O'Hare).
- **Europe:** Also the European subnetwork shows small-world properties, with an average shortest path length of 3,2. This is almost exactly proportional by $k=2$ to $\log(n)$ (6,4). In this case, the top three hubs (and authorities) are BCN (Barcelona), AMS (Amsterdam) and FRA (Frankfurt).
- **USA:** As expected, the U.S. network follows small-world behavior. The most significant hubs and authorities include ATL (Atlanta) and ORD (Chicago), as seen in the global context, followed by LAX (Los Angeles).
- **Africa:** Also in this case we can notice small-world properties, with the most important airports being NBO (Nairobi), JNB (Johannesburg) and ADD (Addis Abeba-Bole). For authorities, instead, the third place is occupied by the ABJ (Félix Houphouët Boigny International) airport, located on the Ivory Coast.

6 Conclusion

Across degree, eigenvector, closeness, and betweenness, the weighted analyses confirm that a small number of “super-hub” airports handle disproportionate traffic and connectivity. These nodes shape both efficiency and fragility: while they enable rapid, high-volume travel, they also create vulnerability if disrupted. Regional patterns—such as Africa’s outliers or the United States’ strong core—highlight how local infrastructure and route distribution affect network structure. Ultimately, the evidence of scale-free or heavy-tailed behavior underscores the importance of planning and resilience measures that account for these highly centralized flows.

Assortative mixing proved to be a good proxy for political closeness between countries: a possible development would be to normalize it by the geographical distance between regions in order to “isolate” the motivation of social proximity between countries for higher assortivity values. The observations on the properties of the 4 networks during the removal of the main hubs, as well as the measurement of the relationship between the average local clustering coefficient and node degrees, confirmed the fact that observing air traffic can provide interesting insights into the social structure and infrastructure development policies of a territory. For example, the differences between Africa and wealthier geographical areas were clearly highlighted. It was interesting to note also the differences between networks developed by multiple political entities (Europe, World) which are by their nature more decentralized, and those designed by a single central government, more “efficient” but more vulnerable, such as the Usa example.

The network and its subnetworks follow expected patterns, with high connectivity in Europe and the USA and significantly lower connectivity in Africa. Since air travel is essential for tourism and trade, it is important to assess whether poorly connected regions are genuinely insignificant or if they play a specialized role in global tourism, emerging economies, or local business networks. Having now identified the weaker links, it is important to understand if they represent lack of demand or latent potential. Understanding the factors behind low connectivity could pave the way for more inclusive air travel networks, fostering greater accessibility and economic growth in underserved regions.

7 Critique

The project addresses the research goals by identifying key hubs and clusters through centrality measures and community detection, confirming the air traffic network's scale-free properties and hierarchical structure typical of transport systems. It also substantiates the link between flight patterns and geopolitical structures, particularly contrasting Europe's politically fragmented network with the Usa's centralized system, and highlighting the efficiency-flexibility trade-off. One limitations was the reliance on static topological data, that did not addressed temporal aspects of the domain: future work could integrate flight frequency, delays and capacity expressed by multilayer network models to better capture resilience and optimize resource allocation.

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